

Ethan Gates

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SUMMARY

AI-Native Software Engineer building production AI products, agent systems, and data-intensive full-stack applications. Healthcare SaaS engineer with hands-on work in grounded AI, local inference, orchestration, and evaluation.

EDUCATION

B.S. Computer Science, Western Governors University Completed: Feb 2025

PROFESSIONAL EXPERIENCE

Software Engineer, DOVAXIS (Healthcare SaaS patient referrals & admissions) May 2025 – Present

Began as a paid engineering intern; converted to full-time November 2025.

- Owned a multi-tenant authorization migration end to end: consolidated scattered role, route, and resource checks into a Permit.io-backed facade with bulk decisions and cached permissions, built the reconciliation report and per-organization readiness verdict that gated the cutover, then ran the production cutover of every customer organization.
- Built and operate a multi-provider LLM service (GraphQL on Cloud Run) that internal services call instead of the OpenAI SDK: a model registry with provider auto-detection across OpenAI and Gemini, a unified quality tier, streaming and tool calling, per-call telemetry for tokens, cost, latency, and organization, plus a gated public chat (pre-launch) with cited web search.
- Built the version and index governance for a multi-tenant clinical document library: a unified version-history and DocHub lifecycle, re-indexing that keeps the AI search index in step with the library, a per-organization vector audit that finds and clears stale entries, and cited-source attribution so answers separate the documents they used from other hits.
- Created the regulation library as a multi-tenant system of record: a master library client organizations subscribe to, a diff-and-sync service that flags new, updated, or removed items while preserving client customizations, read-only import into each client's document tree, and selected regulations wired into vector search so client AI answers cite them.
- Rebuilt how a legacy healthcare app proves changes are safe, with agentic workflows generating and governing the suites: an offline Firebase emulator with sanitized fixtures so tests stop writing to production, browser and Firebase-backed suites wired into a co-designed CI/CD pipeline, LLM-run audits that replay past production bugs and surfaced nine live ones, four patched the same week, and type checking and lint behind shrink-only baselines, 113k dead lines removed.

INDEPENDENT PROJECTS

- GatesAI Chat:** Built and shipped a local-first AI workspace (React/Tauri/Rust with a Go bridge): OpenRouter/Ollama routing, multi-round tool execution, sandboxed files/shell/Git/SQLite, memory, and image generation, with automated checks for streams, tools, and 100+ user journeys. [\[Repo\]](#)
- Agent Handles:** Built a toolkit that makes a web app addressable by AI agents: a Vite plugin verifies a stable identity on every interactive element, a registry maps each to its source line, journeys as data compiled to Playwright, and a drive API that runs the app with a receipt per action. Under it: an explorer that crawls the app and files failures as replayable journeys, an English-to-action resolver with confidence gating, and a harness that runs one agent turn per receipt. [\[Demo\]](#)
- Fathom:** Built a market-analysis suite in TypeScript and React: portfolio backtesting over 155 years of asset-class history, Trinity-validated Monte Carlo retirement simulation, SEC-filing research, and broker-CSV portfolio X-ray. [\[Demo\]](#)
- Media Vault:** Built a private photo and video library that runs on your own machine: timeline, people, places, trips, and search, with local models for faces, text in photos, image search by description, and video transcripts. [\[Repo\]](#)
- Milo:** Built a local voice companion: whisper.cpp, Ollama, and Pocket TTS in one Python process, interruptible mid-sentence, first audio about 300 ms after you stop talking, and an animated robot whose mouth tracks playback. [\[Repo\]](#)

TECHNICAL SKILLS

- Languages:** TypeScript, Python, Rust, JavaScript, SQL, C#.
- AI / LLM Engineering:** Agent orchestration, context management, tool calling, grounded retrieval, embeddings and vector search, evaluation and benchmarking, cost and latency telemetry, OpenAI, Gemini, OpenRouter, Ollama.
- Frameworks & Libraries:** React, MobX, FastAPI, Tauri, TensorFlow, NumPy, Pandas, Tailwind CSS.
- Tools & Infrastructure:** Git/GitHub, Playwright, Firebase, PostgreSQL, Docker, Vite, Cloud Run, Linux.